GMF Investments - Portfolio Optimization Report

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Executive Summary

This report presents the results of our time series forecasting and portfolio optimization analysis. We compared ARIMA and LSTM models for Tesla stock prediction, optimized portfolio allocation using Modern Portfolio Theory, and backtested the strategy against a 60/40 benchmark.

Key Findings:

- LSTM model outperformed ARIMA by -0.1% in accuracy
- Max Sharpe portfolio delivered 35.3% vs benchmark 11.7%
- Strategy outperformed benchmark by 23.6 percentage points
- Conservative portfolio maintains 85.6% in bonds for capital preservation

1. Model Performance Analysis

Model	MAE	RMSE	MAPE (%)
ARIMA	74.80	81.18	32.7
LSTM	74.73	81.10	32.7
Improvement	0.1%	0.1%	0.1%

Selected Model: LSTM (Lower values indicate better performance)

2. Portfolio Recommendations

Max Sharpe Portfolio (Aggressive Growth Strategy):

TSLA	65.2%
BND	0.0%
SPY	34.8%

Minimum Volatility Portfolio (Conservative Strategy):

TSLA	5.3%
BND	85.6%
SPY	9.1%

3. Backtest Performance (Aug 2024 - Jul 2025)

Metric	Strategy	Benchmark (60/40)	Difference

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Total Return	**35.3%**	**11.7%**	**+23.6%**
Annualized Return	**63.1%**	**13.5%**	**+49.6%**
Annual Volatility	**52.1%**	**12.4%**	**+39.6%**
Sharpe Ratio	**0.94**	**1.02**	**-0.08**

4. Risk Assessment & Recommendations

- The aggressive portfolio offers significant growth potential but comes with higher volatility
- The conservative portfolio provides stability with 85.6% allocation to bonds
- Strategy volatility (52.1%) is significantly higher than benchmark (12.4%)
- For risk-averse clients: Recommend Minimum Volatility portfolio
- For growth-oriented clients: Recommend Max Sharpe portfolio with proper risk disclosure

5. Investment Recommendations

- Implement dynamic rebalancing based on monthly forecast updates
- Consider adding stop-loss mechanisms for the aggressive portfolio
- Monitor Tesla's fundamental indicators alongside technical forecasts
- For new allocations, consider dollar-cost averaging into the strategy
- Regular review of model performance and recalibration as needed